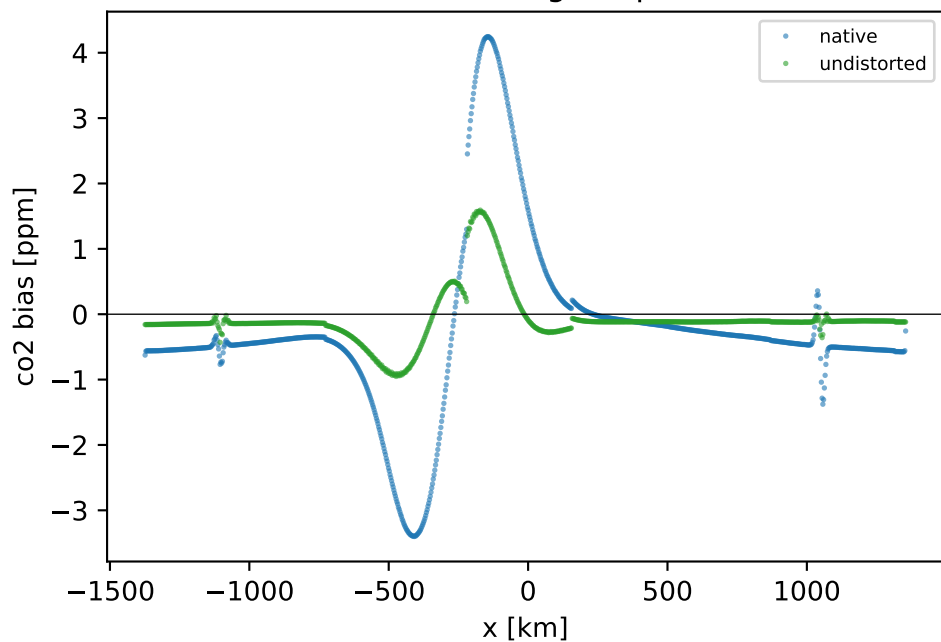
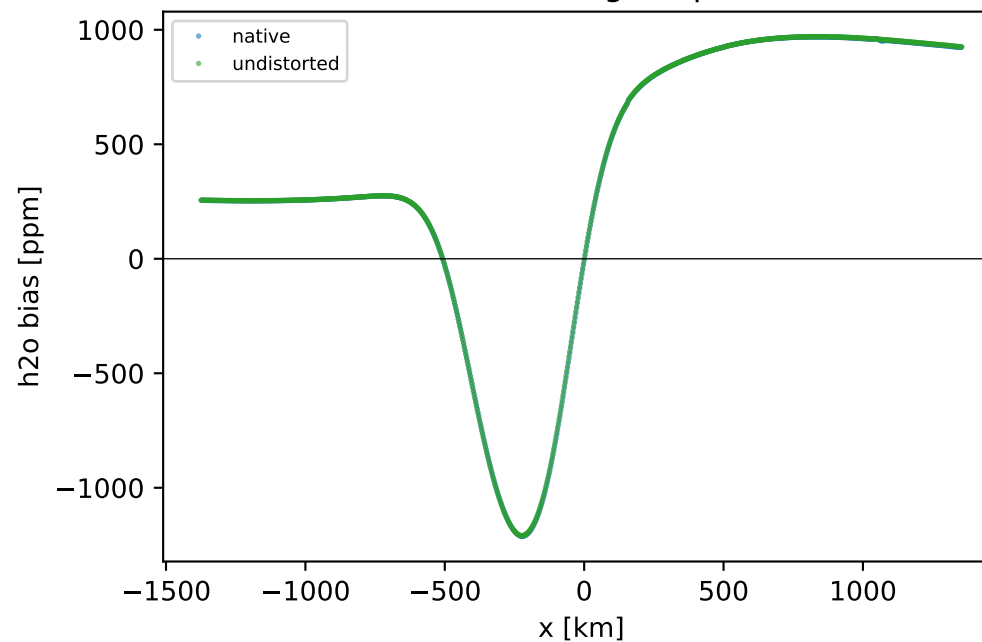


Joint FPA0+FPA2 -- realistic, no noise (native/undistorted)

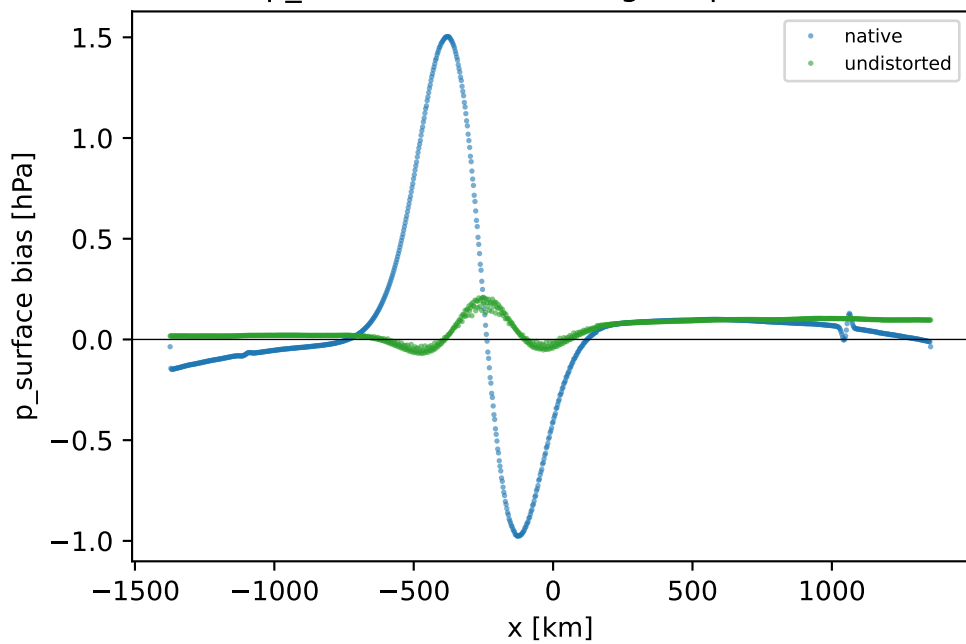
co2 bias vs along-slit position



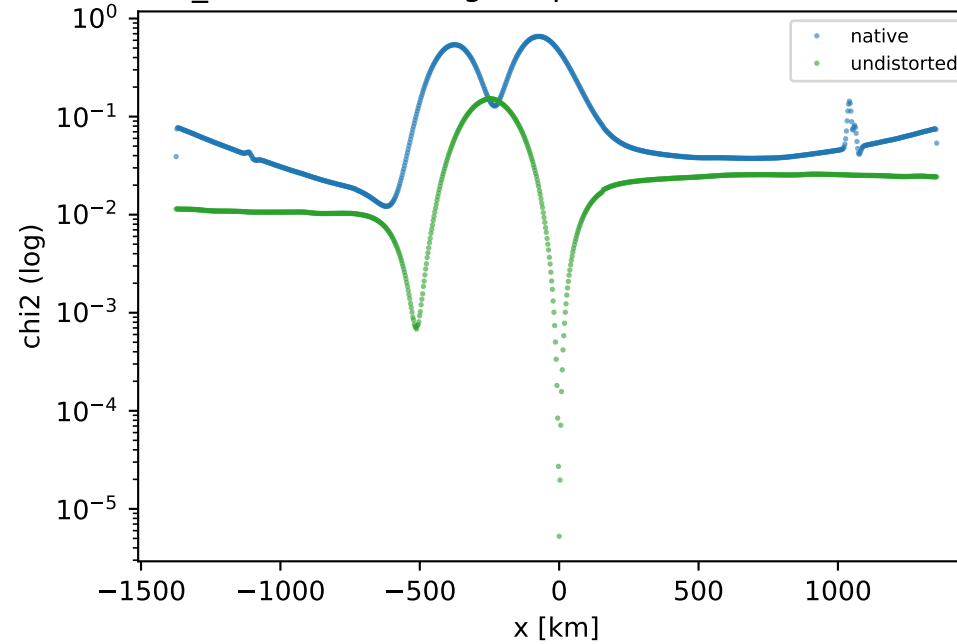
h2o bias vs along-slit position



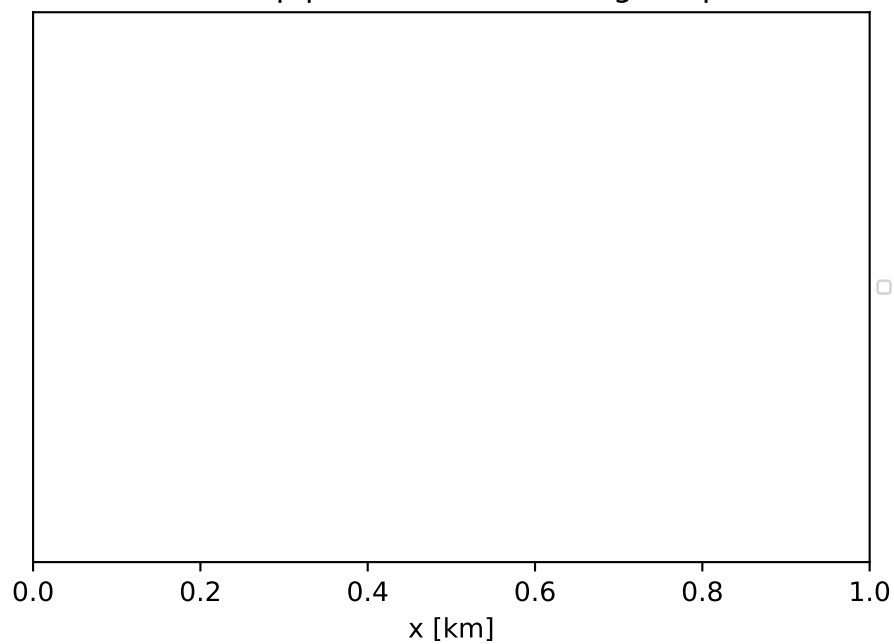
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

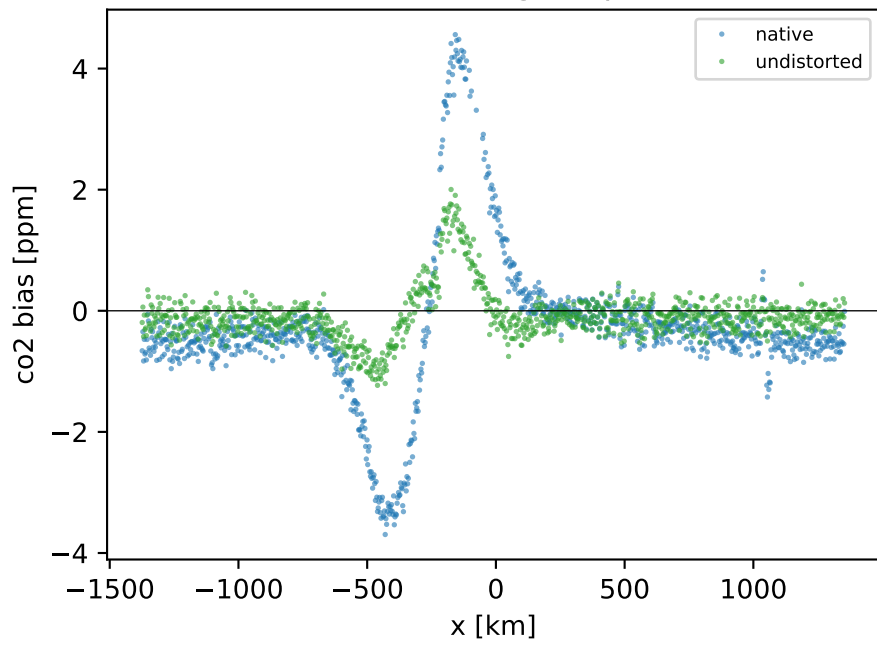


Where each pipeline fails, vs along-slit position

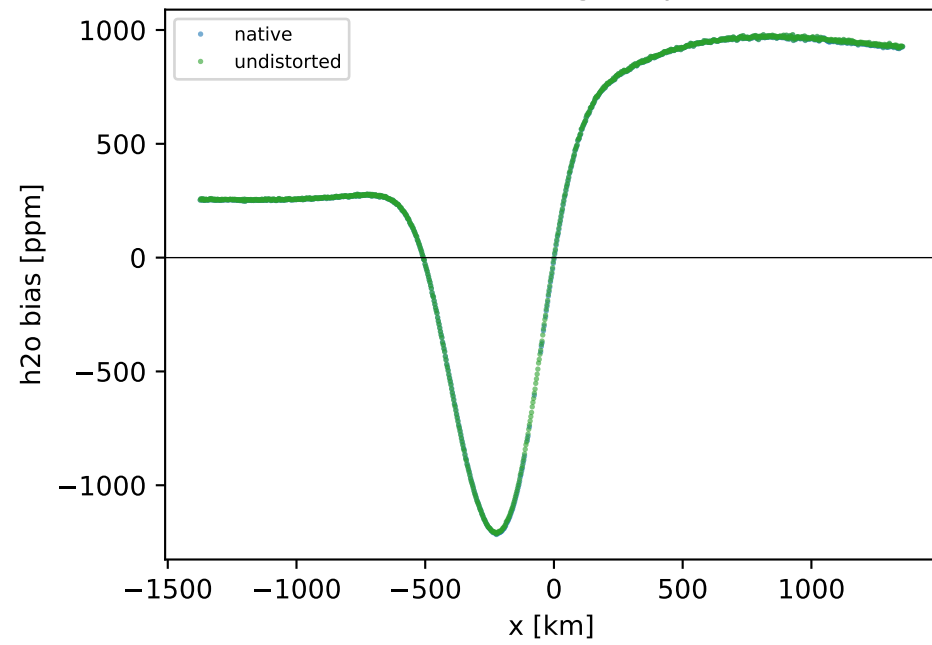


Joint FPA0+FPA2 -- realistic, noise (native/undistorted)

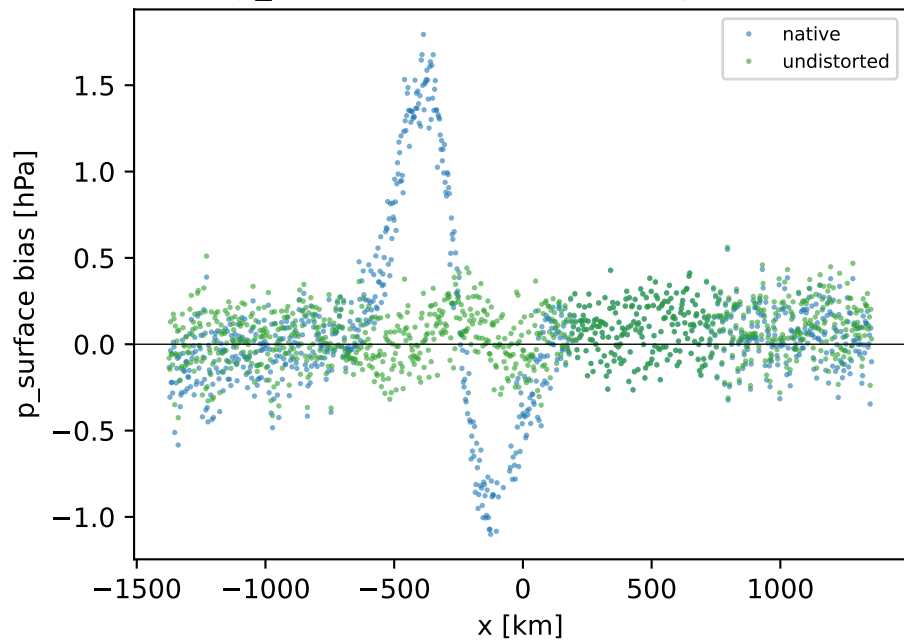
co2 bias vs along-slit position



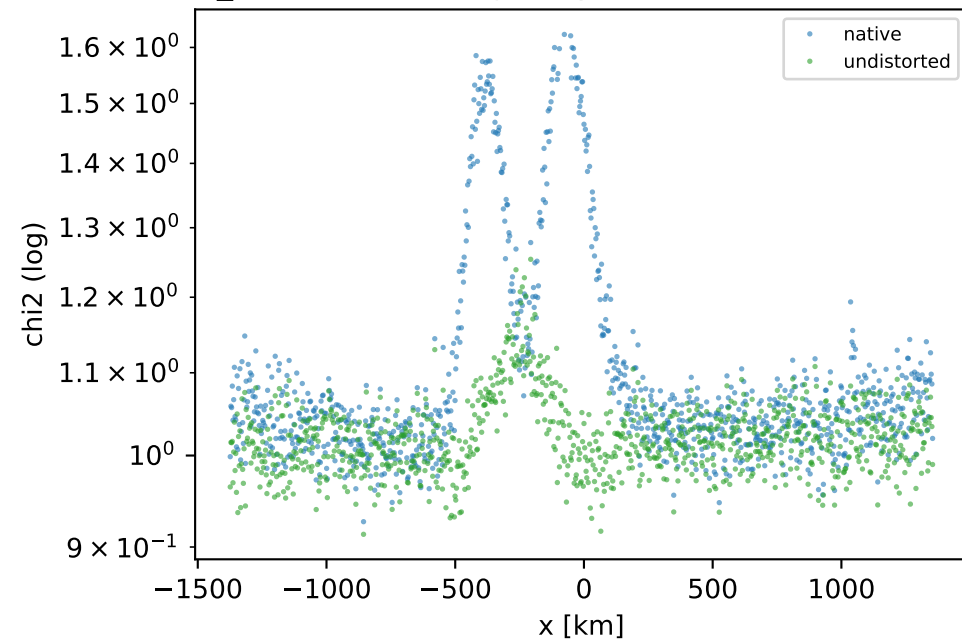
h2o bias vs along-slit position



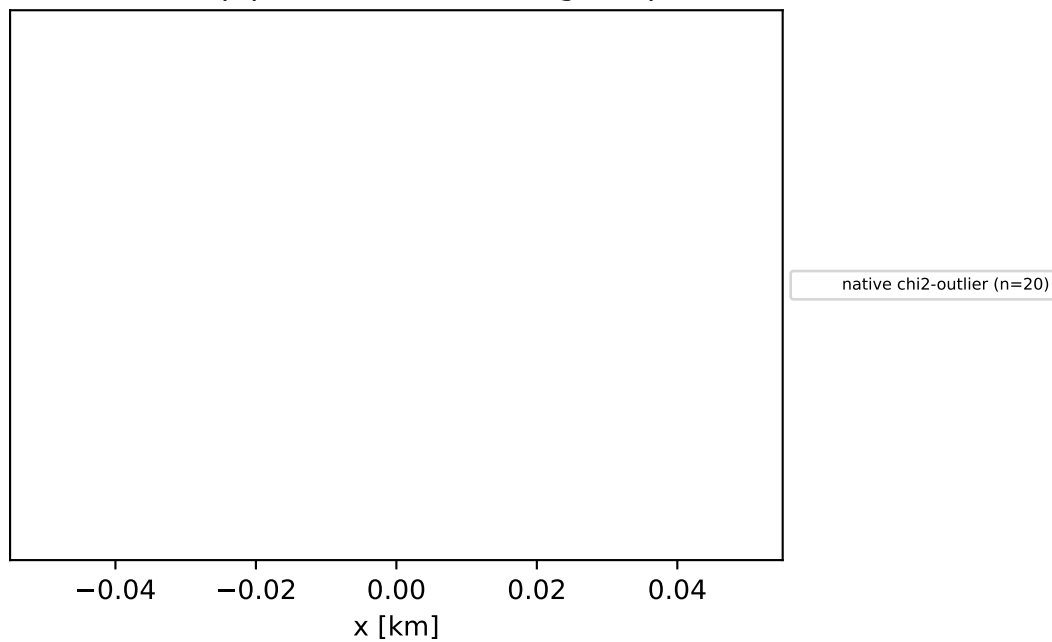
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

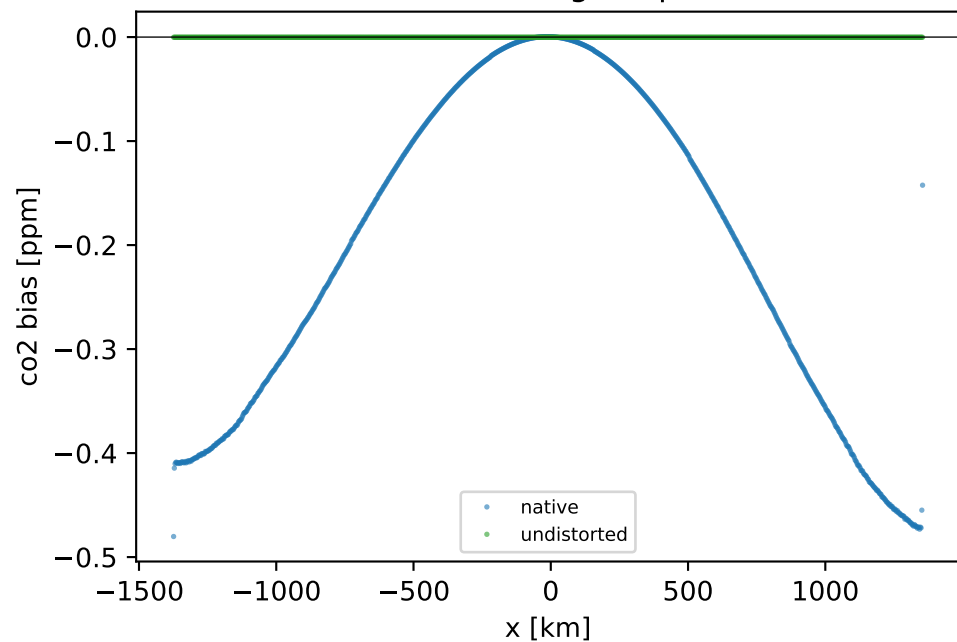


Where each pipeline fails, vs along-slit position

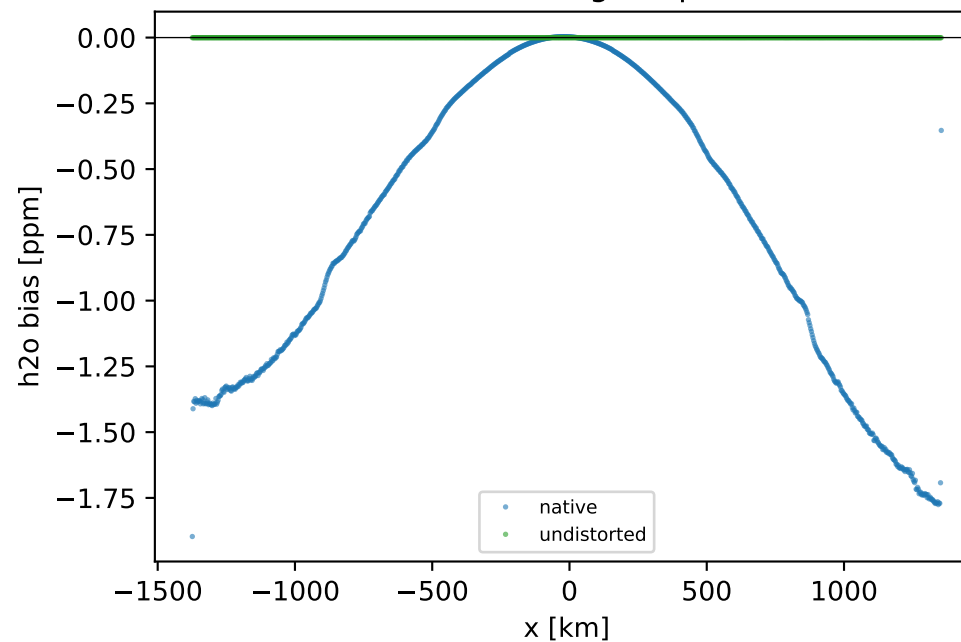


Joint FPA0+FPA2 -- uniform, no noise (native/undistorted)

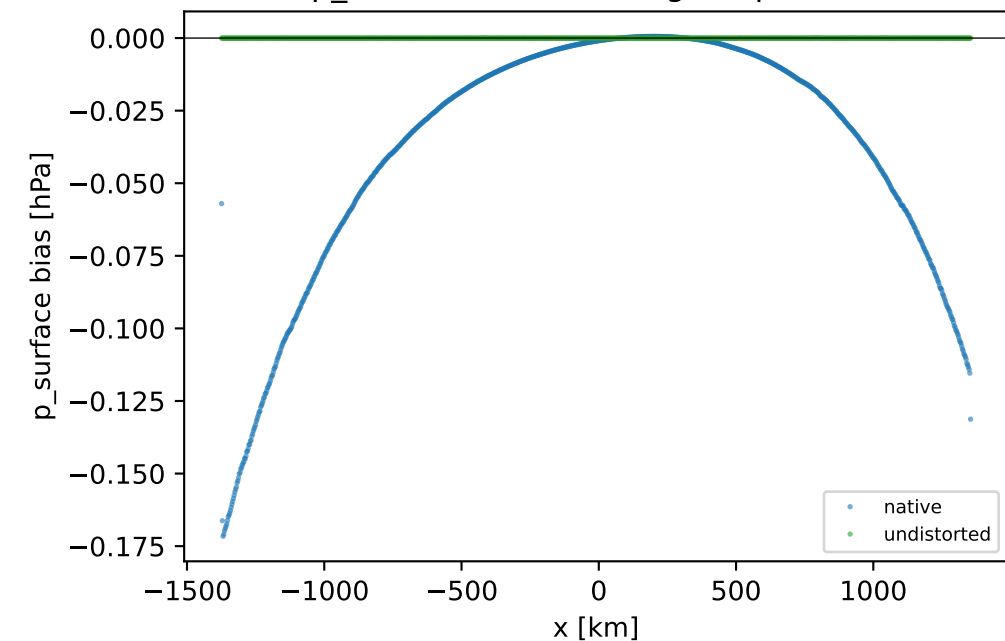
co2 bias vs along-slit position



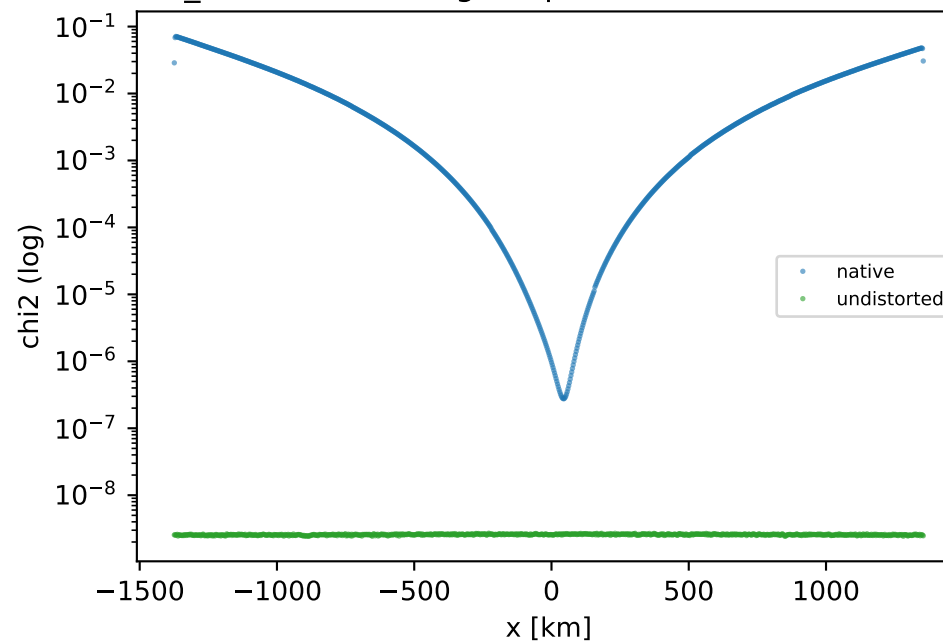
h2o bias vs along-slit position



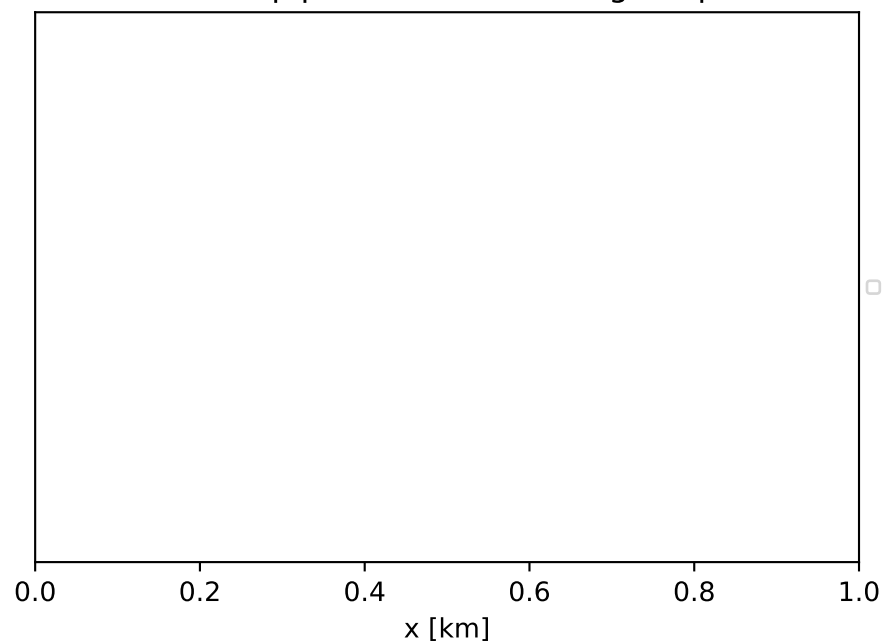
p_surface bias vs along-slit position



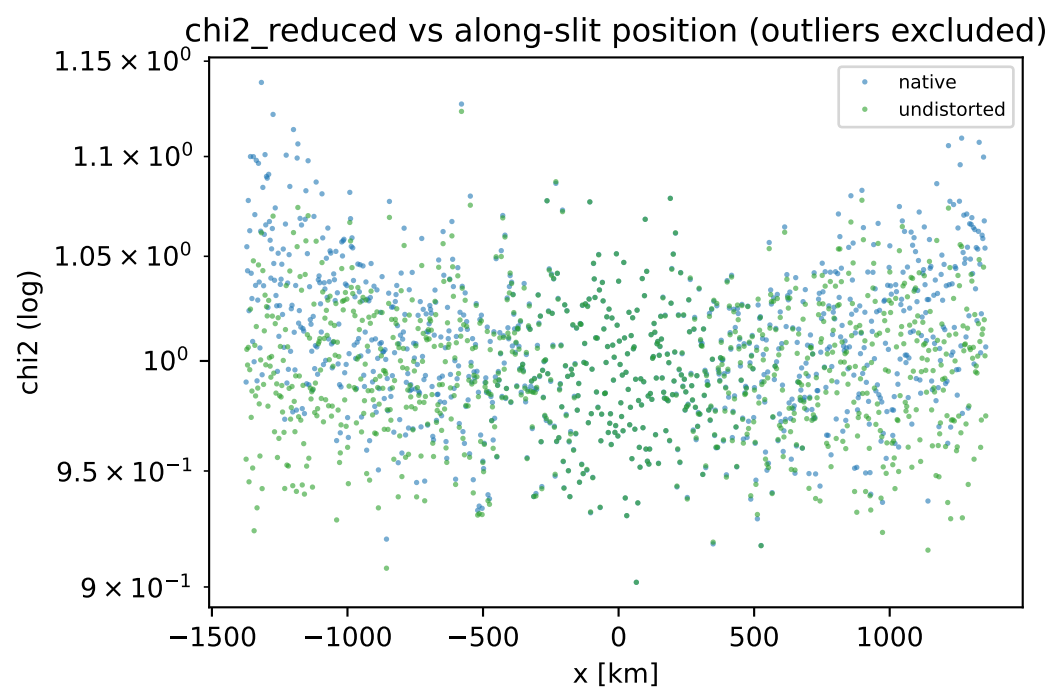
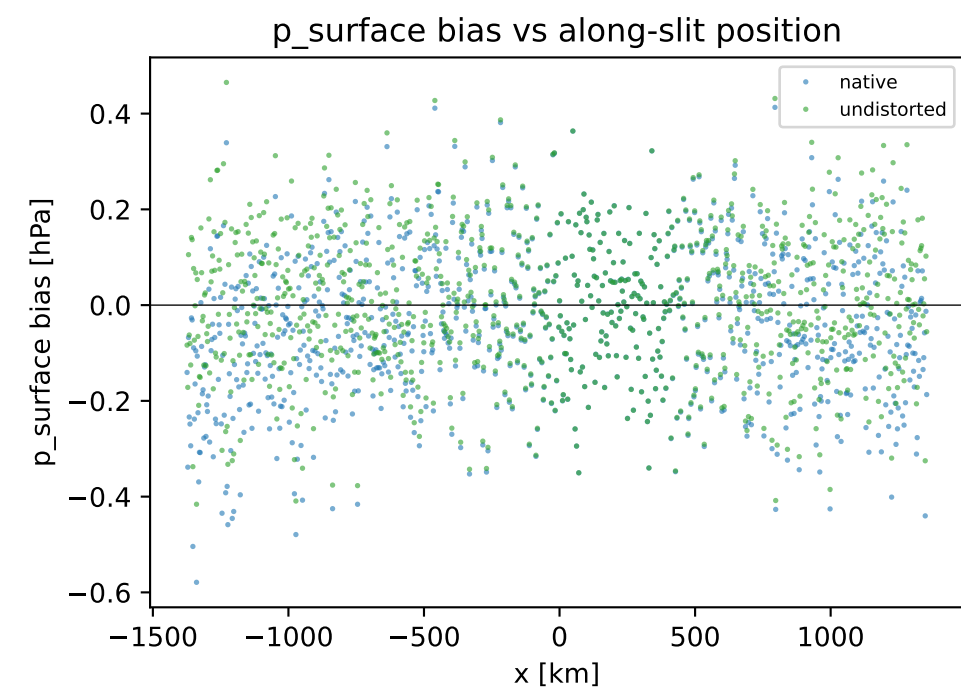
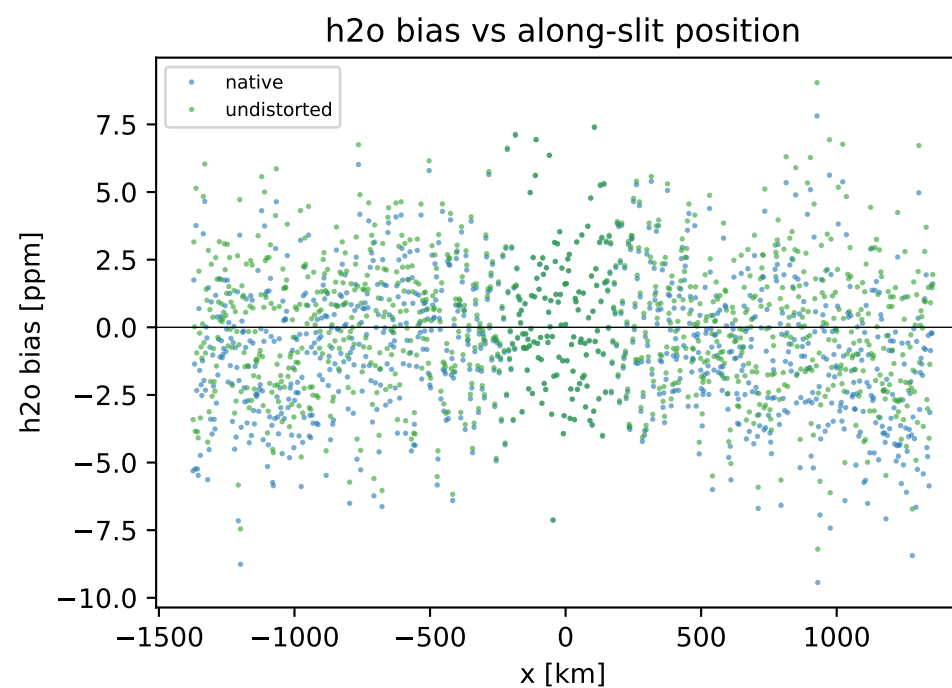
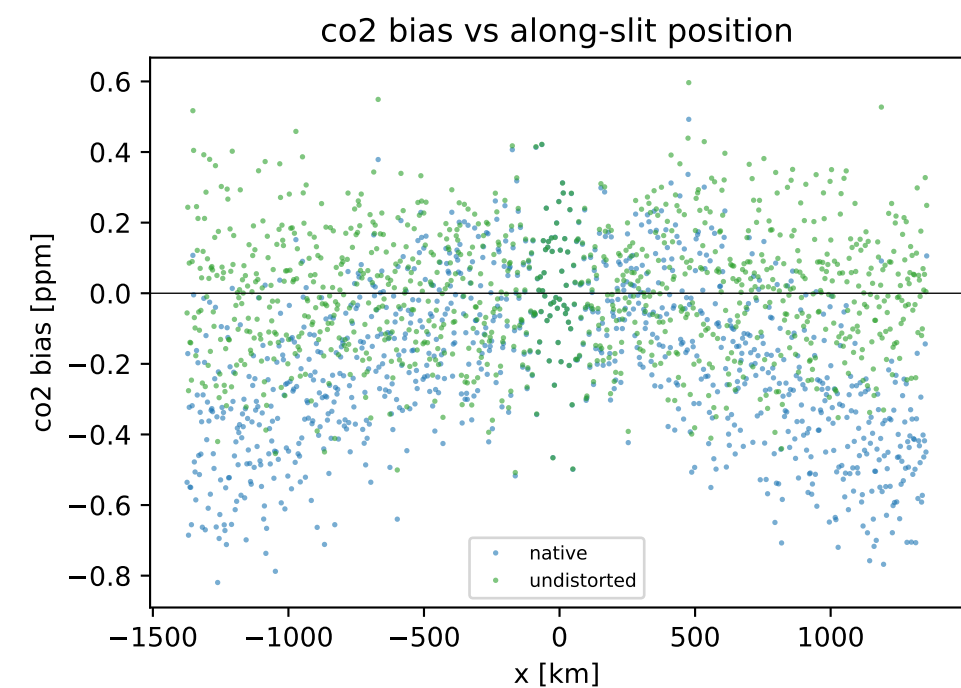
chi2_reduced vs along-slit position (outliers excluded)



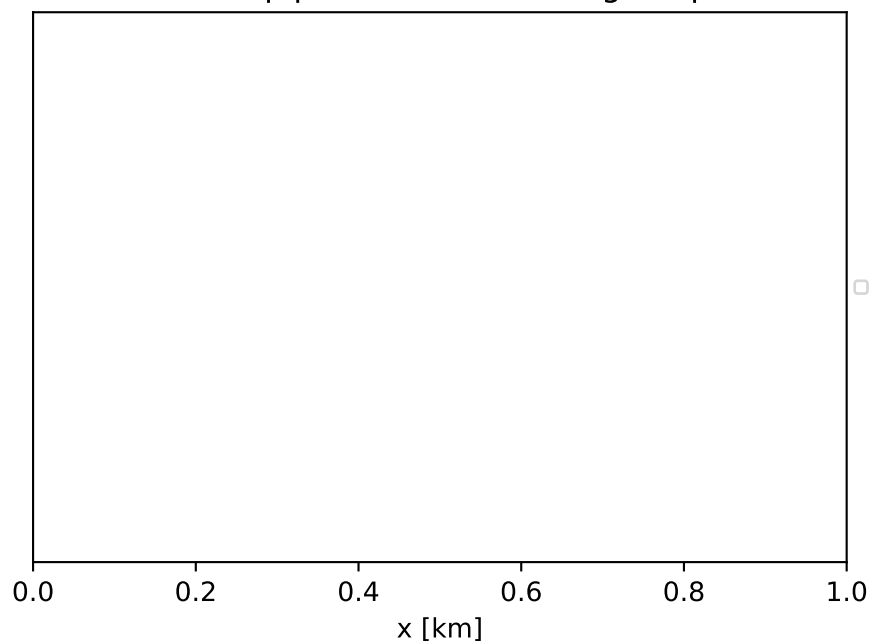
Where each pipeline fails, vs along-slit position



Joint FPA0+FPA2 -- uniform, noise (native/undistorted)

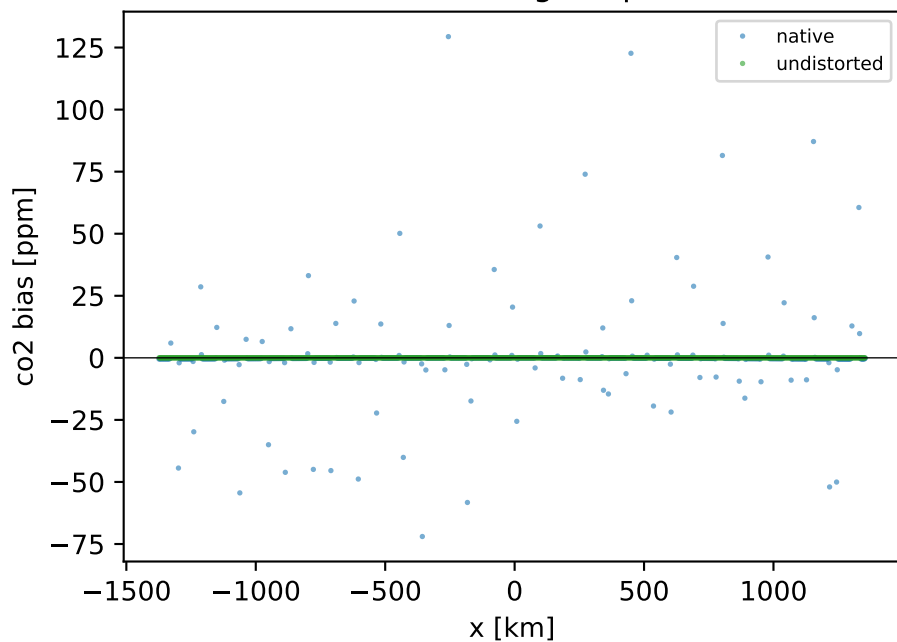


Where each pipeline fails, vs along-slit position

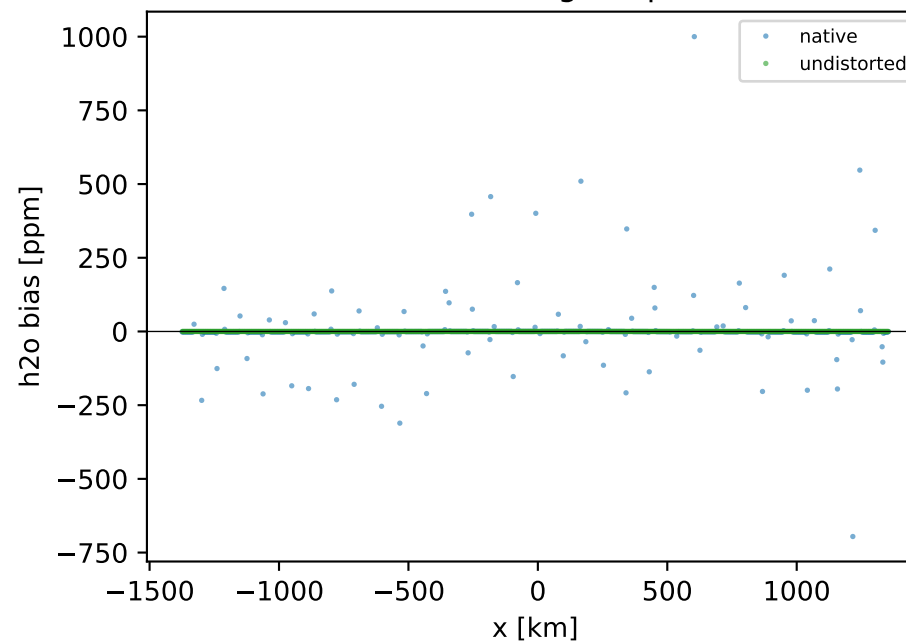


Joint FPA0+FPA2 -- barcode, no noise (native/undistorted)

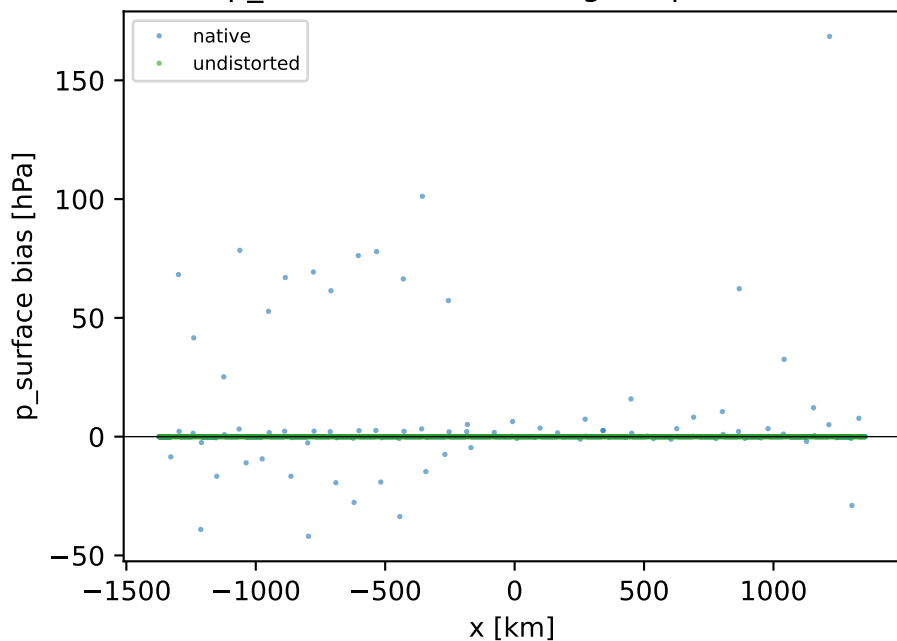
co2 bias vs along-slit position



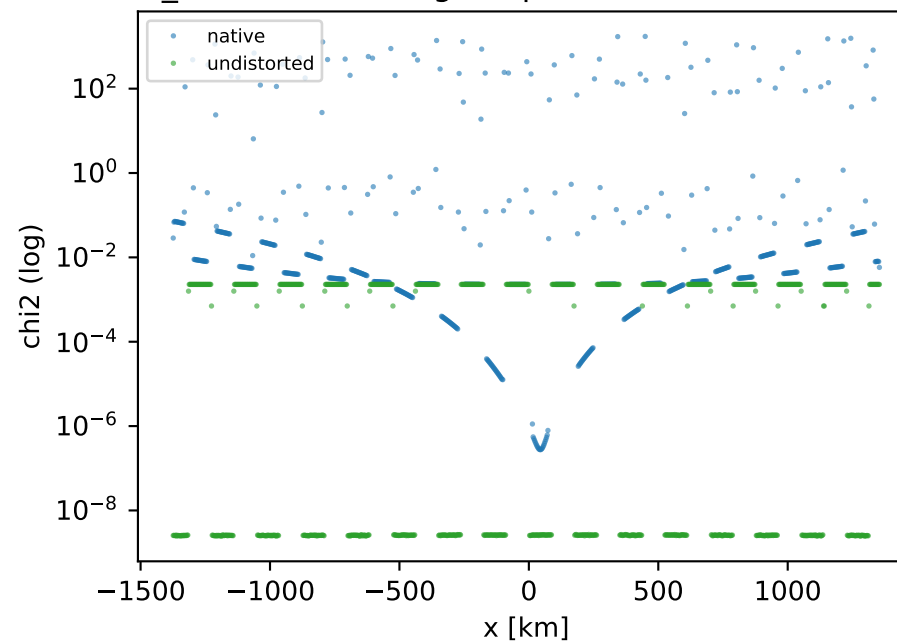
h2o bias vs along-slit position



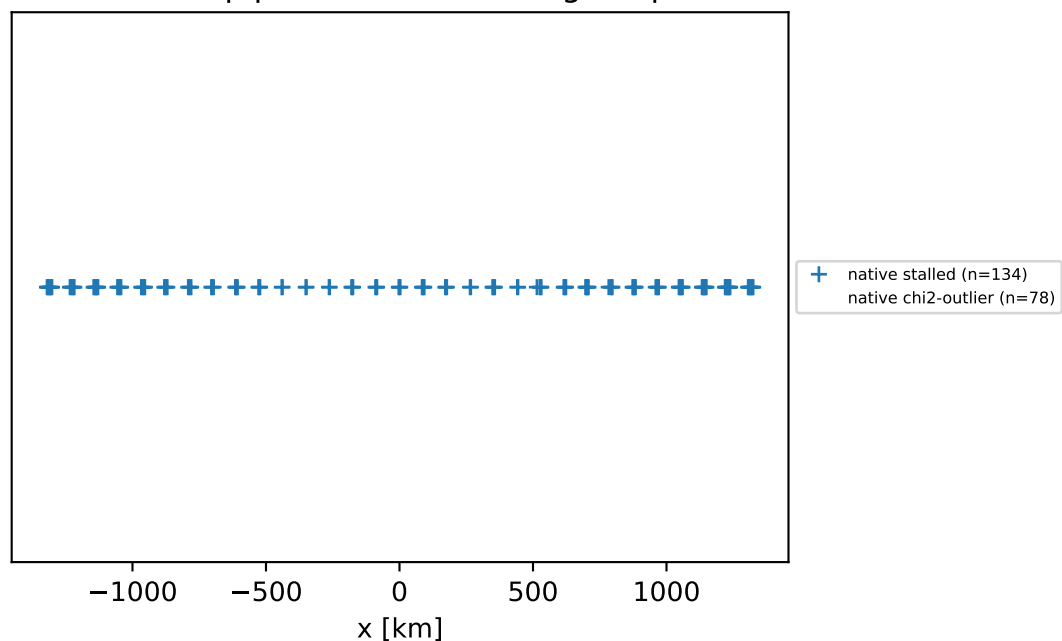
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

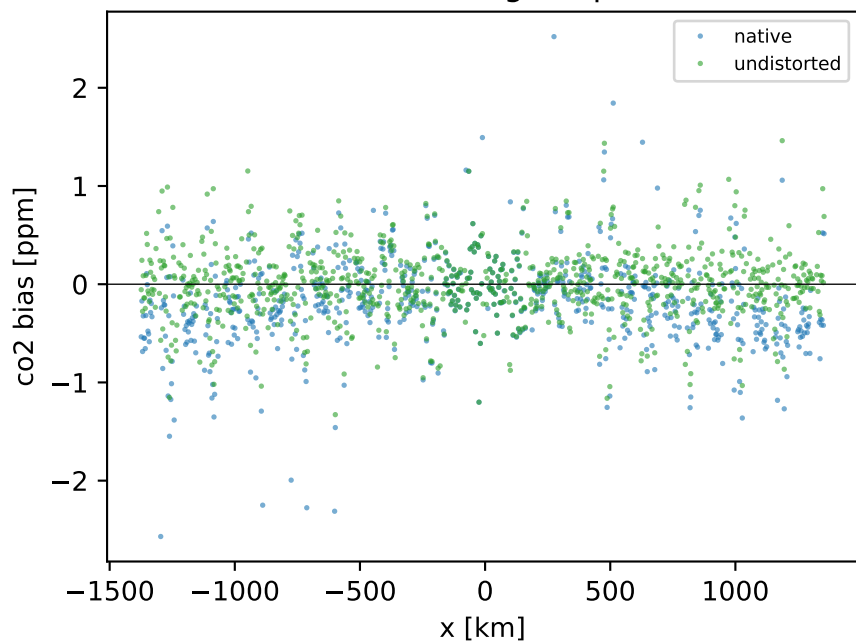


Where each pipeline fails, vs along-slit position

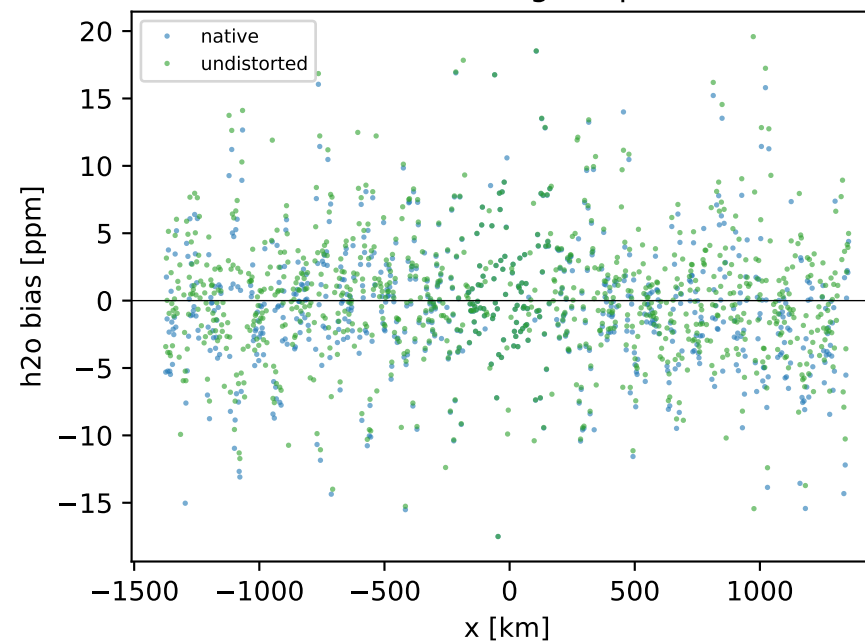


Joint FPA0+FPA2 -- barcode, noise (native/undistorted)

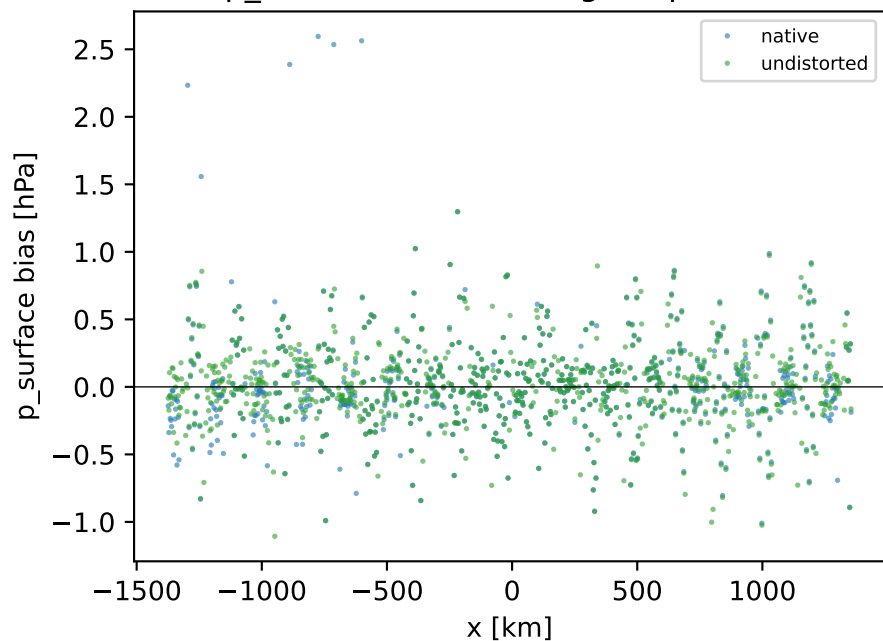
co2 bias vs along-slit position



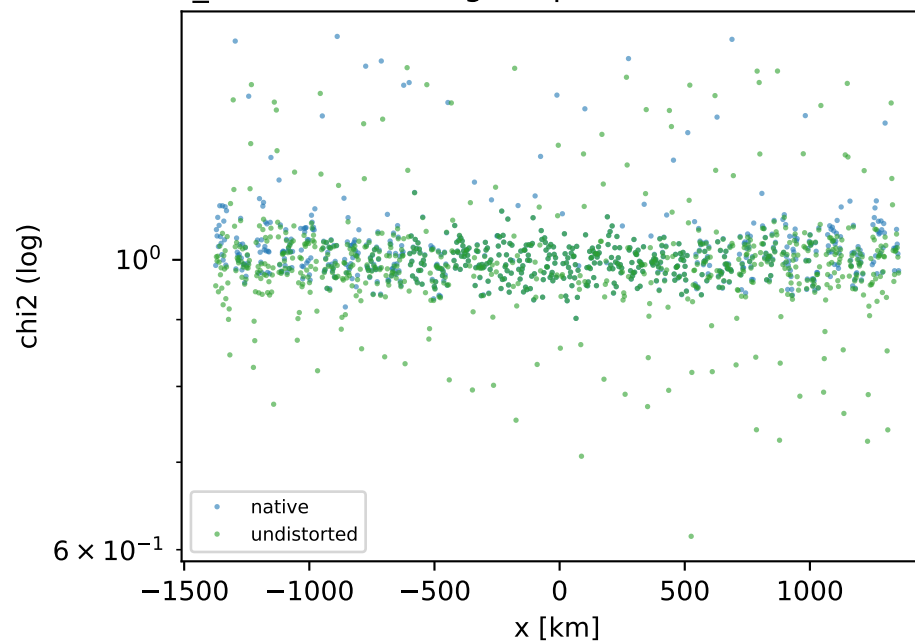
h2o bias vs along-slit position



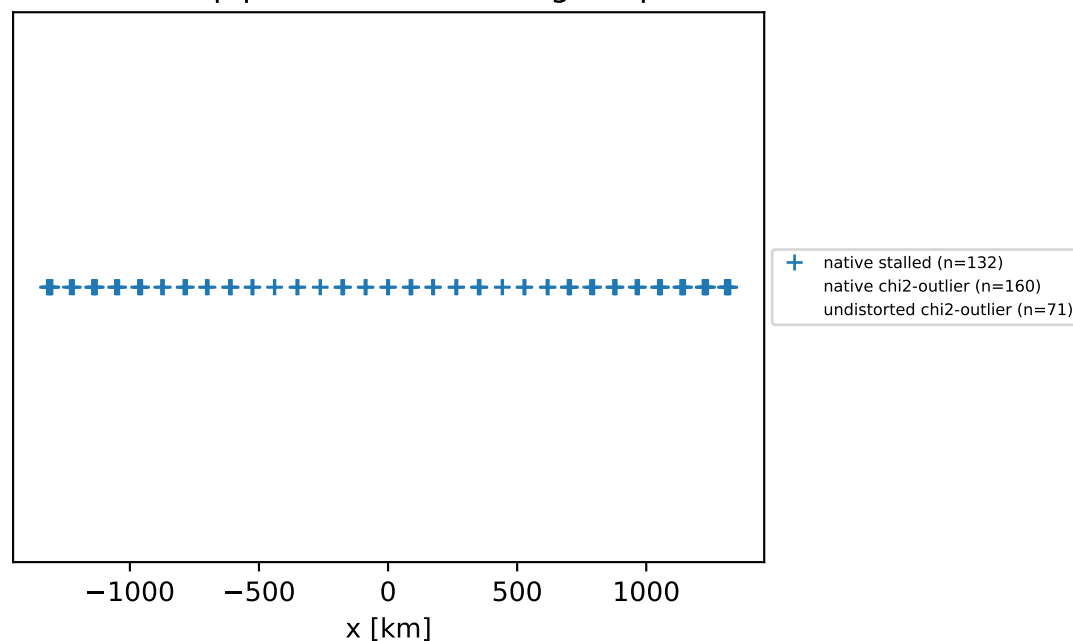
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA0+FPA2 -- robust bias statistics across scenes/noise (native/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	native	1007	-0.2349	1.323	-0.3968	0.2394	144
realistic	0	h2o	native	1007	+370.1	628.5	+273.7	899.9	0
realistic	0	p_surface	native	1007	+0.07229	0.4303	+0.04768	0.09198	129
realistic	0	co2	undistorted	1007	-0.08471	0.3971	-0.1179	0.02975	208
realistic	0	h2o	undistorted	1007	+371.6	628.2	+274.2	900.4	0
realistic	0	p_surface	undistorted	1007	+0.05551	0.05396	+0.07167	0.06819	0
realistic	1	co2	native	987	-0.3038	1.247	-0.3778	0.3961	42
realistic	1	h2o	native	987	+389.8	618.7	+275.5	881.5	0
realistic	1	p_surface	native	987	+0.096	0.4429	+0.04226	0.227	0
realistic	1	co2	undistorted	1007	-0.08208	0.4319	-0.1182	0.2451	2
realistic	1	h2o	undistorted	1007	+371.6	628.1	+274.5	901	0
realistic	1	p_surface	undistorted	1007	+0.06001	0.1581	+0.06465	0.1698	0
uniform	0	co2	native	1007	-0.2018	0.1527	-0.1878	0.2146	0
uniform	0	h2o	native	1007	-0.7249	0.5549	-0.6622	0.7789	0
uniform	0	p_surface	native	1007	-0.03835	0.0424	-0.02125	0.02997	0
uniform	0	co2	undistorted	1007	-0.0001906	3.038e-06	-0.0001906	3.225e-06	0
uniform	0	h2o	undistorted	1007	-0.0004732	3.772e-05	-0.0004747	3.771e-05	0
uniform	0	p_surface	undistorted	1007	-2.937e-05	7.268e-06	-3.113e-05	6.872e-06	0
uniform	1	co2	native	1007	-0.1995	0.2355	-0.1959	0.2484	0
uniform	1	h2o	native	1007	-0.6807	2.591	-0.6403	2.626	0
uniform	1	p_surface	native	1007	-0.03389	0.1522	-0.03015	0.1523	0
uniform	1	co2	undistorted	1007	+0.002159	0.1795	+0.004255	0.1839	0
uniform	1	h2o	undistorted	1007	+0.04365	2.48	+0.01964	2.415	0
uniform	1	p_surface	undistorted	1007	+0.004466	0.1458	+0.004795	0.1573	0
barcode	0	co2	native	795	+0.09983	11.59	-0.1634	0.2221	76
barcode	0	h2o	native	795	+1.371	69.61	-0.5733	0.8085	88
barcode	0	p_surface	native	795	+1.145	11.21	-0.0006447	0.01632	115
barcode	0	co2	undistorted	1007	-0.0001424	5.136e-05	-0.000183	0.0001241	0
barcode	0	h2o	undistorted	1007	-0.0003141	0.0001785	-0.0003819	0.0002391	0
barcode	0	p_surface	undistorted	1007	-4.602e-06	2.701e-05	-7.223e-06	3.768e-05	0
barcode	1	co2	native	715	-0.1769	0.4632	-0.1805	0.3393	0
barcode	1	h2o	native	715	-0.4933	4.861	-0.5564	3.585	0
barcode	1	p_surface	native	715	+0.02368	0.3755	-0.01247	0.2285	5
barcode	1	co2	undistorted	936	+0.004272	0.3538	+0.005834	0.2686	0
barcode	1	h2o	undistorted	936	+0.2172	4.738	+0.03426	3.579	0
barcode	1	p_surface	undistorted	936	+0.01634	0.3	+0.009653	0.2197	0